

SURENDRA NADH SOMALA

304, Dept. of Civil Engg., IIT Hyderabad, Kandi, Sangareddy, Hyderabad surendra@ce.iith.ac.in

Research Interests

AI/ML, Heritage Structures

Employment

Associate Professor, Indian Institute of Technology Hyderabad (IITH), 2021 - present

Assistant Professor, Indian Institute of Technology Hyderabad (IITH), 2015 - 2021

Senior Scientist – Physics, Pinnacle-Halliburton, 2014 - 2015

Educational Background

California Institute of Technology (Caltech)	Ph.D. in Civil Engineering (geophysics & seismology)	2009-2013
California Institute of Technology (Caltech)	M.S. in Civil Engineering	2008-2009
Indian Institute of Technology (IIT) Guwahati	B.Tech in Civil Engineering	2004-2008

Scholarship and Award

Stanford 2% list, 2025

Indian National Science Academy (INSA) Young Associate Award (IYA), 2024

Faculty Research Excellence Award (IITH), 2022

Keck Institute of Space Studies Graduate Fellowship, 2012

George W. Housner Graduate Fellowship in Earthquake Engineering, 2008

Synergistic activities

Member of LIGO Scientific Collaboration (LSC)

Life Member of Indian Society of Earthquake Technology (ISET)

Co-developer of a Finite Element code (PyLith) for dynamic elastic and quasi-static viscoelastic problems

Co-developer of Spectral Element code (SPECFEM3D) for 3D wave propagation and rupture dynamics

Projects (Role: *PI*)

DRDO (2024) Generation of Data Cube Module in GeoTiff Format [1 year] *Role: PI*

MoES (2023) Seismic Source Characterization using State-of-the-art Signal Processing and Machine Learning Techniques [3 years] *Role: PI*

DST (2021) [International] OrigamiSat for Structural Health Monitoring [2 years] *Role: PI*

ISRO (2020) Strain Estimation in Central Seismic Gap using Numerical Modeling constrained by Geodetic Measurements [3 years] *Role: PI*

DST-RFBR (2019) A comparative study between the Great Caucasus (Russia) and the Himalaya [2 years] *Role: PI*

IUCAA (2018) Seismic studies for LIGO-India site to reduce low frequency noise [3 years] *Role: PI*

MoES (2017) Dynamic Slip Inversion for Earthquakes with Available Data [3 years] *Role: PI*

SERB-ECR (2016) Investigating Earthquake Source Physics Inclusion into Engineering Analysis [3 years] *Role: PI*

International Conferences and Workshops (selected)

4th International Conference on Earthquake Early Warning, Seoul, South Korea, September, 2019

16th World Conference on Earthquake Engineering (16WCEE), Santiago, Chile January, 2017

International Natural Hazards and Infrastructure (ICONHIC2016), Crete Island, Greece, June 2016

International Continental Drilling Program (ICDP) DSeis workshop, South Africa, November, 2015

(after ASP) IITH affiliated Publications with IITH student

Saha, S., K. Katyal, and **S. N. Somala** (2025) Deep Learning-Based Imputation Framework in Bridge Health Monitoring using Generative Adversarial Networks, *Knowledge-Based Systems*, 311, 113088 (doi: 10.1016/j.knosys.2025.113088)

Parla, R., S. Veggalam, and **S. N. Somala** (2025) Collapse fragility of structures subjected to basin amplified ground motion: Influence of basin parameters, *Advances in Structural Engineering*, (<https://doi.org/10.1177/13694332251348616>)

Saha, S., K. S. K. Karthik Reddy, K. Katyal and **S. N. Somala** (2024) Investigating Seismic Performance of SMRF Buildings under Subshear and Supershear Rupture Conditions: Assessment of Storey-level Internal Damage, *Engineering Structures*, 309, 118069 (doi: 10.1016/j.engstruct.2024.118069)

Payyappilly L.J., K. S. K. Karthik Reddy, and **S. N. Somala** (2024) Influence of Rupture Velocity on Risk Assessment of Concrete Moment Frames: Supershear Vs. Subshear Ruptures, *Structures*, 60, 105895 (doi: 10.1016/j.istruc.2024.105895)

Somala, S. N., S. Chanda, M. AlHamaydeh, and S. Mangalathu (2024) Explainable XBoost-SHAP Machine-Learning Model for Prediction of Ground Motion Duration in New Zealand, *ASCE Natural Hazards Review*, 25(2), 04024005

Karthik Reddy, K. S. K., S. Veggalam, and **S. N. Somala** (2022) Spatial Variation of Structural Fragility due to Supershear Earthquakes, *Structures*, 44, 389-403 (doi: 10.1016/j.istruc.2022.08.025)

Chaudhari, V., K. S. K. Karthik Reddy, and **S. N. Somala** (2022) Offshore Wind Turbine Subjected to Supershear Earthquake Ruptures, *Advances in Structural Engineering*, 25(15), 3072-3085 (doi:10.1177/13694332221115466)

Chaudhari, V., and **S. N. Somala** (2022) Fragility of Offshore Wind Turbines Variation with Pulse-period and Amplitude: Directivity and Fling step, *Structures*, 41, 66-76 (doi: 10.1016/j.istruc.2022.04.078)

Somala, S. N., S. Mangalathu, S. Chanda, K. S. K. Karthik Reddy, and R. Parla (2022) Focal Mechanism Influence with Azimuth using Near-Field Simulated Ground Motion: Application to Multi-Span Continuous Concrete Single Frame Box Girder Bridge, *ASCE Journal of Bridge Engineering*, 27(6), 04022034 (doi: 10.1061/(ASCE)BE.1943-5592.0001875)

Somala, S. N., R. Parla, and S. Mangalathu (2022) Basin Effects on Tall Bridges in Seattle from M9 Cascadia, *Engineering Structures*, 260, 114252 (doi: 10.1016/j.engstruct.2022.114252)

Somala, S. N., K. S. K. Karthik Reddy, and S. Mangalathu (2022) Diaphragm Abutment Californian Bridges Subjected to UCERF2 Rupture Scenarios: Complete Damage State Evolution with Improvements to Seismic Codes, *Soil Dynamics and Earthquake Engineering*, 155, 107204 (doi: 10.1016/j.soildyn.2022.107204)

Mandal, P., and **S.N. Somala**, (2022) Functionally Graded Materials Pile Structure for Seismic Noise Cancellation, *ASCE International Journal of Geomechanics*, 22(10), 04022168 (doi: 10.1061/(ASCE)GM.1943-5622.0002537)

Saha, S., K. S. K. Karthik Reddy, and **S. N. Somala** (2022) Seismic Assessment of Steel Frame Subjected to Simulated Directivity Earthquakes: The Unilaterality of Fault Normal Component at Various Rupture Distances, *Journal of Building Engineering*, 47, 103880 (doi: 10.1016/j.jobe.2021.103880)

Dwivedi, A., K. S. K. Karthik Reddy, and **S. N. Somala** (2022) Study of Seismic Orientation of Structure with Bi-directional Response Analysis in the Vicinity of Branched Fault Earthquake Rupture, *Structures*, 37, 613-623 (doi: 10.1016/j.istruc.2022.01.027)

Parla, R., and **S. N. Somala** (2022) Seismic Ground Motion Amplification in 3D Sedimentary Basins: The Effect of

Source Mechanism and Intensity Measures, *Journal of Earthquake and Tsunami*, 16(4), 2250008 (doi: 10.1142/S1793431122500087)

Somala, S. N., K. Karthikeyan, and S. Mangalathu (2021) Time Period Estimation of Masonry Infilled RC Frames using Machine Learning, *Structures*, 34, 1560-1566 (doi: 10.1016/j.istruc.2021.08.088)

Somala, S. N., K. S. K. Karthik Reddy, and S. Mangalathu (2021) The Effect of Rupture Directivity, Distance and Skew Angle on the Collapse Fragilities of Bridges, *Bulletin of Earthquake Engineering*, 19(14), 5843-5869 (doi: 10.1007/s10518-021-01208-8)

Somala, S. N., S. Chanda, K. Karthikeyan, and S. Mangalathu (2021) Explainable Machine Learning on New Zealand Strong Motion for PGV and PGA, *Structures*, 34, 4977-4985 (doi: 10.1016/j.istruc.2021.10.085)

Somala, S.N., S. Chanda, M.C. Raghucharan, and E. Rogozhin (2021) Spectral Acceleration Prediction for Strike, Dip, Rake: A Multi-layered Perceptron Approach, *Journal of Seismology*, 25(5), 1339-1346 (doi: 10.1007/s10950-021-10031-2)

Chaudhari, V., and **S. N. Somala** (2021) Seismic performance of offshore wind turbine in the vicinity of seamount subduction zone, *Structures*, 34, 423-432 (doi: 10.1016/j.istruc.2021.07.080)

Karthik Reddy K.S.K., **S. N. Somala**, and H.H. Tsang (2021) Response of Inelastic SDOF Systems Subjected to Dynamic Rupture Simulations of Directivity and Fling Step, *Soil Dynamics and Earthquake Engineering*, 151, 106992 (doi: 10.1016/j.soildyn.2021.106992)

Veggalam, S., K. S. K. Karthik Reddy, and **S. N. Somala** (2021) Collapse Fragility due to Near-field Directivity Ground Motions: Influence of Component, Rupture distance, Hypocenter location, *Structures*, 34, 3684-3702 (doi: 10.1016/j.istruc.2021.09.096)

(after ASP) IITH affiliated Publications without IITH student

Somala, S.N. (2021) Lockdowns and their Influence on Earth's Hum, *Scientific Reports* 11, 17838 (doi: 10.1038/s41598-021-97459-1)
<https://www.nature.com/articles/s41598-021-97459-1>

(before ASP) IITH affiliated Publications with IITH student

Chanda, S., and **S. N. Somala** (2021) Single Component Single Station based Machine Learning for estimating Magnitude and Location of an Earthquake: A Support Vector Machine approach, *Pure and Applied Geophysics*, 178(6), 1959-1976 (doi: 10.1007/s00024-021-02745-8)

Somala, S.N. , S. Chanda, M.C. Raghucharan, and E. Rogozhin (2021) Spectral Acceleration Prediction for Strike, Dip, Rake: A Multi-layered Perceptron Approach, *Journal of Seismology* (doi: 10.1007/s10950-021-10031-2)

Raghucharan, M. C., **Somala, S. N.**, Erteleva, O., and Rogozhin, E. (2021) *Seismic attenuation model for data gap regions using recorded and simulated ground motions*. *Natural Hazards*, 107, 423-446, doi: 10.1007/s11069-021-04589-w

Raghucharan, M C and **Somala, S N** and Rodina, Svetlana (2019) *Seismic attenuation model using artificial neural networks*. *Soil Dynamics and Earthquake Engineering*, 126. p. 105828. ISSN 02677261

Raghucharan, M C and **Somala, S N** (2018) *Seismic damage and loss estimation for central Indo-Gangetic Plains, India*. *Natural Hazards*. ISSN 0921-030X

Raghucharan, M C and **Somala, S N** (2017) *Simulation of strong ground motion for the 25 April 2015 Nepal (Gorkha) Mw 7.8 earthquake using the SCEC broadband platform*. *Journal of Seismology*, 21 (4). pp. 777-808. ISSN 1383-4649

(before ASP) IITH affiliated Publications without IITH students

Parameswaran, R.M., K. Rajendran, **S. N. Somala**, and C. P. Rajendran (2020) *The 2012 Mw 8.6 Indian Ocean earthquake: Deep nucleation on a listric-like fault*. *Physics of Earth and Planetary Interiors*, 307, 106550, doi: 10.1016/j.pepi.2020.106550

Mukund, N., B. O'Reilly, **S. N. Somala**, and S. Mitra (2019) *Effect of Induced Seismicity on Advanced Gravitational Wave Interferometers*. *Classical and Quantum gravity* 36(10), 10LT01, doi: 10.1088/1361-6382/ab1360

Somala S. N., J. P. Ampuero, & N. Lapusta (2018) *Finite-fault source inversion using adjoint methods in 3-D heterogeneous media*. *Geophysical Journal International*, 214(1), 402-420. doi: 10.1093/gji/ggy148

Harms, J., J.-P. Ampuero, M. Barsuglia, E. Chassande-Mottin, J.-P. Montagner, **S. N. Somala**, B. Whiting (2015), *Transient gravity perturbations induced by earthquake rupture*, *Geophysical Journal International*, 201 (3), 1416-1425 doi:10.1093/gji/ggv090

Harris, R. A., M. Barall, B. Aagaard, S. Ma, D. Roten, K. Olsen, B. Duan, D. Liu, B. Luo, K. Bai, J.-P. Ampuero, Y. Kaneko, A.-A. Gabriel, K. Duru, T. Ulrich, S. Wollherr, Z. Shi, E. Dunham, S. Bydlon, Z. Zhang, X. Chen, **S. N. Somala**, C. Pelties, J. Tago, V. M. Cruz-Atienza, J. Kozdon, E. Daub, K. Aslam, Y. Kase, K. Withers, L. Dalguer (2018) *A Suite of Exercises for Verifying Dynamic Earthquake Rupture Codes*. *Seismological Research Letters*, vol. 89 (3), 1146-1162, doi: <https://doi.org/10.1785/0220170222>

Mai, P.M., D. Schorlemmer, M. Page, J.P. Ampuero, K. Asano, M. Causse, S. Custodio, W. Fan, G. Festa, M. Galis, F. Gallovic, W. Imperatori, M. K  ser, D. Malyskyy, R. Okuwaki, F. Pollitz, L. Passone, H. Razafindrakoto, H. Sekiguchi, S. Song, **S. N. Somala**, K. Thingbaijam, C. Twardzik, M. van Driel, J. Vyas, R. Wang, Y. Yagi, and O. Zielke (2016) *The earthquake-source inversion validation (SIV) project*, *Seismological Research Letters*, 87 (3) 690-708; doi: 10.1785/0220150231

Prior to IITH Publications

Somala S. N., J. P. Ampuero, & N. Lapusta (2018) *Finite-fault source inversion using adjoint methods in 3-D heterogeneous media*. *Geophysical Journal International*, 214(1), 402-420. doi: 10.1093/gji/ggy148

Somala, S. N., J.-P. Ampuero, N. Lapusta (2014), *Resolution of rise time in earthquake slip inversions: Effect of station spacing and rupture velocity*, *Bulletin of Seismological Society of America*, 104 (6). pp. 2717-2734, ISSN 0037-1106, doi: 10.1785/0120130185

Galvez, P., J.-P. Ampuero, L. Dalguer, **S. N. Somala**, T. Nissen-Meyer (2014), *Dynamic earthquake rupture modeled with an unstructured 3D spectral element method applied to the 2011 M9 Tohoku earthquake*, *Geophysical Journal International*, 198 (2), 1222-1240, doi:10.1093/gji/ggu203.

Michel, R., J.-P. Ampuero, J.-P. Avouac, N. Lapusta, S. Leprince, D.C. Redding, **S. Somala**, (2013), *A Geostationary Optical Seismometer, Proof of Concept*, *IEEE Transactions on Geoscience and Remote Sensing*, 51 (1): 695–703. doi:10.1109/TGRS.2012.2201487.

Raghu Kanth, S.T.G, Lakshmi Kumari, K., **S. N. Somala** (2012), *Regional level ground motion simulation for a hypothetical great earthquake in the Garwhal Himalaya*, *Geomatics and Natural Hazards* 0 (0): 1–24. doi:10.1080/19475705.2012.731658.

Raghu Kanth, S.T.G, **S. N. Somala** (2009), *Modeling of Strong-Motion Data in Northeastern India: Q , Stress Drop, and Site Amplification*, *Bulletin of the Seismological Society of America* April 2009 vol. 99 no. 2A 705-725doi: 10.1785/0120080025